

MV-STGNN Ablations & Significance

Ablation Study Framework (04_ablations_mvstgnn.ipynb)

Protocol: The main model (A0) gets retrained for 90 epochs, every time, using a 5-fold cross-repetition setup. This way, nothing sneaks in from early stopping or training differences.

3-Tier Registry (25 variants):

- **Tier 1 (Core Impact):** Drops graphs (A1), SupCon loss (A9), or Subject FiLM (P1) to test if primary innovations drive performance.
- **Tier 2 (Attribution):** Isolates specific graph views, removes synergy pooling, tests dense attention, and swaps GroupNorm for BatchNorm.
- **Tier 3 (Sensitivity):** Sweeps synergy counts (K), model width, and window/stride lengths.

Paired Thresholding: Computes $\Delta = A0 - \text{variant}$ (balanced accuracy per subject) to mechanically classify component utility without subjective bias.

Significance Testing (05_significance.ipynb)

Integrity Gate: Enforces complete paired-score matrices and matches MD5 split fingerprints across all models before running tests.

Statistical Pipeline:

Omnibus Gate: Runs a Friedman test ($\alpha = 0.05$); post-hoc tests abort if the omnibus result is non-significant to prevent multiplicity abuse.

Post-Hoc Testing: Evaluates MV-STGNN against baselines (Family F1) and ablations (Family F2) using two-sided Wilcoxon signed-rank tests with Holm-Bonferroni corrections, Cohen's d_z , and BCa bootstrap confidence intervals.

Secondary Analysis: Produces Nemenyi critical-difference diagrams and optional linear mixed-effects models as confirmatory cross-checks.